

# CodeTrackr — Short Interview Guide

A plain-English summary of the full 64-page prep document. Read this once or twice and you can hold your own in an interview. Everything here is how the code actually works, not how the project was described.

## October 2026 update — read this first (added 2026-10-03)

This document was written before the October 2026 roadmap (`docs/ROADMAP_2026-10.md`). That work is **built and tested but not deployed yet**, and extension **2.5.0 is not published** — say "I built", not "it's live". **Wherever the text below disagrees with this box, this box wins.** Up-to-date short versions: `CodeTrackr_Interview_Cheat_Sheet.md` (rewritten) and `docs/INTERVIEW_PREP.md`. Measured numbers: `docs/BENCHMARKS.md`.

| Topic                | Old text says                                                          | Now                                                                                                                                                                                                                                                                                         |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extension API key    | plaintext, non-expiring, unscoped; in <code>settings.json</code>       | <code>ct_&lt;id&gt;_&lt;secret&gt;</code> , DB stores SHA-256 of the secret (id lookup + <code>timingSafeEqual</code> ), shown once; <b>device-code sign-in</b> issues per-device keys that expire in 1 year and can be revoked one by one; extension keeps the key in <b>SecretStorage</b> |
| Leaderboard          | $O(\text{all activity} + \text{all users})$ scan per request (H-7/H-8) | <code>userstats</code> running totals updated on every write; all-time board = 3 indexed reads. <b>1M rows: p50 6.72 s → 62 ms</b> (local benchmark). Group boards too; contest windows ? <code>from=&amp;to=</code>                                                                        |
| Cheating             | a valid key + loop inflates hours (H-21)                               | credited time = $\min(\text{duration}, \text{focus} + 120 \text{ s})$ , $\leq 600 \text{ s}$ per user per 10-minute window via an atomic counter, long uploads spread; per-key quota; raw claim kept                                                                                        |
| Duplicates / offline | no idempotency; memory-only buffer, lost on restart                    | <code>flushId</code> receipts (applied once); persisted outbox in <code>globalState</code> , FIFO, survives restart (M-30 fixed)                                                                                                                                                            |
| Analytics            | <code>find()</code> then JS reduce (M-1)                               | daily/weekly from one <code>\$facet</code> pipeline, proven equal to the old code by an oracle test                                                                                                                                                                                         |
| Write reduction      | "~10× fewer writes"                                                    | <b>measured:</b> 4.9× fewer documents, 12.7× less data at the 2-minute cadence (19.6× / 51× vs the old 30 s); same throughput                                                                                                                                                               |
| Tests                | 18 suites / 296; nothing against a real DB                             | unit <b>23 suites / 346</b> , <b>35 integration tests</b> (supertest + in-memory MongoDB, incl. CORS preflight), extension <b>6 / 70</b> ; no frontend tests                                                                                                                                |

| Topic                     | Old text says                  | Now                                                                                                                                                                       |
|---------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| userId type               | String on activities (M-6)     | ObjectId migration in progress (reads accept both; script converts)                                                                                                       |
| Teams                     | orphaned page + live API       | deleted                                                                                                                                                                   |
| Groups                    | no admin powers                | creator can rename / remove members; ownership passes on when the creator leaves                                                                                          |
| Ops                       | no Docker, no CD, console logs | Dockerfile + CI health check, deploy-on-green (off until a secret), JSON logs with request ids, optional Sentry                                                           |
| Bugs found while building | —                              | <b>H-23</b> CORS lacked PATCH → goal completion and mark-as-read never worked in a browser; M-31 goal window; M-32 weekly window; H-22 IST "today"; email leaks M-28/M-29 |
| Still open                | —                              | H-19 cross-site auth cookie, H-20 IP-keyed rate limits, client-chosen timestamps, no web refresh token                                                                    |

**Changed 2026-09-08:** the extension still flushes counters every ~30s, but the backend now **merges flushes into one record per 10-minute window** (\$inc upsert) instead of one record per flush. It also skips flushes with no real activity (extension 2.3.0). The "one record per flush" / `Activity.create` mentions below are the old model — totals are identical, only time-of-day detail is now 10-minute-grained.

Two labels are used throughout:

- **REAL** = this is in the code right now.
- **BETTER** = this is what you'd do to improve it (say it as a suggestion, not a claim).

## 1. What CodeTrackr is (in one breath)

CodeTrackr watches how you code and shows you the numbers — built so a friend group can keep **friendly competition** going.

**The motive:** my friends and I run coding contests and daily-practice streaks, and everyone always *claims* they put in the hours. CodeTrackr makes it automatic: make a **group**, everyone installs the extension, and the group page shows who actually coded, in what languages, and (because it tracks failed commands/builds too) roughly who fought the most errors.

There are three parts:

1. **A VS Code extension.** It runs quietly in your editor. It notices what file you're in, what language, how much you type, how many terminal commands you run **and whether they passed or failed**, when you commit — and every ~2 minutes it sends a small summary to a server.
2. **A backend server** (Node + Express + MongoDB). It checks who the data belongs to using a personal key, then **merges the summary into a 10-minute "bucket" record** (many summaries → one row),

and does the maths when the website asks for it.

3. **A React website** (the dashboard). You log in with Google. It shows charts of your coding, **groups with a per-member leaderboard** (the main feature), a global leaderboard, goal tracking, and an "Insights" page with things like your most productive two-hour window.

#### Two different logins:

- The **website** uses Google sign-in → the server gives you a JWT stored in a cookie.
- The **extension** uses a **64-character API key** you copy from the website.

**Where it runs:** website on Vercel, server on Render, database on MongoDB Atlas.

#### The one honest thing to say first

The "Insights" page is **not machine learning**. It's ordinary maths — averages, medians, a "how steady are you" score. No model, no training, no AI. That was a deliberate choice so every number can be explained. (More in section 9.)

---

## 2. Three ways to explain it

### 30 seconds

"CodeTrackr is a coding-activity tracker built around friendly competition. My friends and I wanted to see who was actually grinding during contest weeks, so I built a VS Code extension that records what you work on — time per language, editor activity, terminal commands and whether they passed or failed, git commits — and sends it to my backend. You make a group with your friends and the app shows a per-member leaderboard plus a global one, goal tracking, and an insights page. MERN stack plus a TypeScript VS Code extension."

### 1 minute

"The idea came from my friend group — during coding contests we kept arguing about who'd done the work, so I built something to measure it. My VS Code extension hooks into editor, terminal, git and window events and keeps running counters — lines added, 'churn' (code you wrote then deleted), focused minutes, command and build success rates. It sends those to an Express API, checked by a per-user API key. The backend merges each summary into a 10-minute record in MongoDB. The React frontend shows daily/weekly Chart.js graphs; **groups you create with a per-member leaderboard**; a global leaderboard; goals on a calendar with cron-driven reminders; and an insights page that works out confidence-gated productivity metrics using plain statistics."

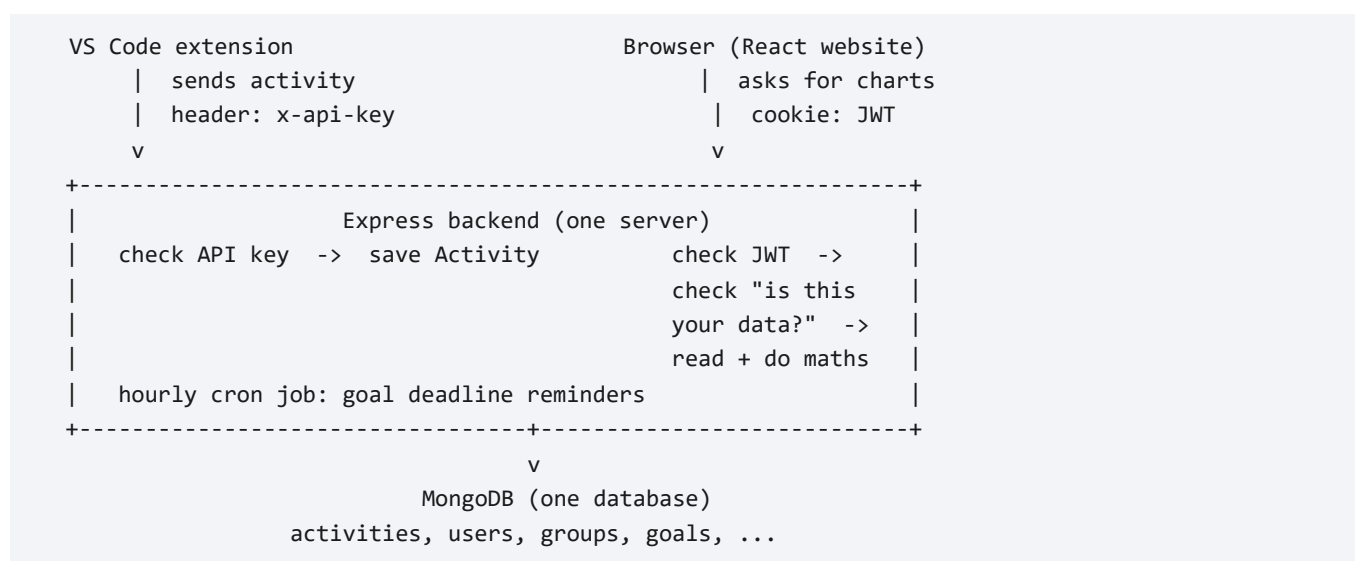
### 2 minutes

Add this to the 1-minute version:

"The interesting design problem was linking the extension to the account without a full login screen inside VS Code. I used an API-key model: on first sign-in the server generates a random 64-character key and stores it on the user. The extension sends it as a header; the server looks up the user by that key. It works, but in a review I'd call it what it actually is — a password-like credential that never expires and is stored as plain text — and I can explain how I'd harden it.

The other thing I'm proud of is the accuracy work. The first version measured time as 'editor was open' and lines as the net change in line count, which is zero for any refactor. I rewrote the trackers to count gross edits and 'churn', and to only count focused time. I also audited my own code into an IMPROVEMENT\_PLAN file with about 30 findings and fixed the serious security ones, with a test that fails if any protected route loses its auth check."

### 3. The big picture (architecture)



**What kind of architecture is this?** Say: "*Client-server, REST-ish, a modular monolith.*"

- **Client-server:** thin clients (extension, website), one server that owns the truth.
- **Monolith:** one Node process. Features are split into separate route files but deploy together. This is fine for one developer and one database.
- **Modular:** each feature (analytics, groups, goals, leaderboard) is its own file, so you could split one out later. The obvious first candidate is the activity-ingest endpoint.
- **No** message queue, **no** Redis cache, **no** WebSockets. Notifications are fetched by the browser every 30 seconds (polling).

**Why not microservices?** One developer, one database, and all the features share the same data. Splitting them would add network calls and complexity for no benefit yet.

## 4. The VS Code extension

### How it starts and runs

When VS Code finishes starting up, the extension wakes (`onStartupFinished`), creates five small "trackers", and starts a timer. The timer ticks every **30 seconds**: if you've been idle 2+ minutes it pauses (resuming when you type again); if you've been active for at least **2 minutes** (the default, `minFlushMinutes` — changed from 30 seconds in v2.3.0) **and** something actually happened (edits / commands / commits), it sends one summary. A flush with no real activity is held so the time carries to the next one.

### What it records

Each tracker keeps counters, then hands them over ("snapshot and reset"):

| Tracker  | What it counts                                                                                                                                |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Editor   | characters and lines typed/deleted, <b>churn</b> (wrote then deleted within 10 min), undo/redo, saves, file switches, time reading vs writing |
| Focus    | minutes the window was actually in front, and "flow blocks" (a work stretch that ends after 2 min idle)                                       |
| Git      | commits (via VS Code's built-in Git, so it catches GUI commits too), uncommitted file count                                                   |
| Terminal | commands run, which succeeded/failed, build/test runs, repeated failures                                                                      |
| Debug    | number of debug sessions                                                                                                                      |

### What it sends (the payload)

One JSON object per flush: timestamp (set to the **start** of the interval, not "now", so hour-of-day charts aren't skewed), file **name only** (never the full path), file type, project, language, duration in **seconds**, line counts, and the four tracker summaries.

**Privacy:** it never sends file contents, diffs, commit messages, or folder paths. Terminal commands are cleaned (`token=...` becomes `token=***`) and shortened.

### What happens when things go wrong

| Situation                | What happens now (REAL)                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| You go idle              | It pauses, then resumes on your next keystroke                                                                                                      |
| VS Code closes           | It tries one last send, then stops                                                                                                                  |
| No internet              | The unsent minutes stay <b>in memory</b> and retry next tick. If VS Code restarts first, that time is lost — <b>there's no on-disk queue</b>        |
| Key is wrong / was reset | You get a one-time popup: "Set API Key" / "Open Dashboard"                                                                                          |
| Same data sent twice     | The backend adds it to the <b>same 10-minute bucket</b> (not a new row), so it double-counts <i>within</i> that bucket — still not fully idempotent |

**Where the key is stored on your machine:** in VS Code's settings file, as **plain text**. **BETTER:** use VS Code's SecretStorage (the OS keychain).

---

## 5. The API key (this is the #1 interview topic)

### How it works (REAL)

1. You sign in with Google for the first time.
2. The server creates your user and runs `crypto.randomBytes(32)` → a 64-character key.
3. It's saved on your user record as **plain text**.
4. The website shows it to you. You paste it into the extension.
5. The extension sends it on every request as a header: `x-api-key: <key>`.
6. The server does `User.findOne({ apiKey })` to find you, then saves the activity under your ID.
7. If you think it leaked, you click "Regenerate" — a new key is made, the old one dies.

### "Is it just an identifier?" — say NO

An **identifier** (like a username) is safe to show publicly. This key is different — holding it lets you **write data**. So it's a **credential**, not an identifier. More precisely:

- It **identifies** you (one key = one user).
- It **authenticates** you (no second check — having it is enough).
- It's a **bearer token** (whoever holds it can use it).
- It **never expires** and has **no limits** on what it can do.
- It's stored and compared as **plain text**.

**Say this line:** *"It's basically a password that never changes, stored in plain text. That's an OK choice for a first version — it's one database lookup and works offline — but I'd call it a credential in a review, and I know how to make it safer."*

### What an attacker could do

With a stolen key they can **fake activity** for you — send fake hours in a loop and mess up the leaderboard and your charts. They **cannot** read your dashboard (that needs your login cookie). There's no rate limit and no sanity check on the numbers, so it's easy.

### How to fix it (BETTER)

- **Store a hash, not the key.** Give the key as `ct_<id>_<secret>`, look up by the `id`, compare the `secret` against a hash. A database leak then reveals nothing usable.
  - **Allow several keys per user**, each named ("laptop", "desktop") and revocable on its own.
  - **Rotation with a grace period** — the old key still works for 24 hours after you make a new one, so tracking doesn't suddenly break.
  - **Even better:** the website hands out a short-lived signed token the extension refreshes, or use an OAuth "device flow" so the extension never holds a long-term secret.
-

## 6. The database (MongoDB)

### The collections

| Collection                   | What it holds                                                                                                  | Notes                                                                                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>activities</b>            | one record per <b>10-minute window</b> (per user + project + language) — changed 2026-09-08; was one per flush | the big one; a 400-day auto-delete rule now caps its growth                                                                                           |
| <b>dailysummaries</b>        | one row per user per day, built by a nightly job from <b>activities</b>                                        | infrastructure for future "all-time" reads — nothing uses it yet                                                                                      |
| <b>users</b>                 | Google ID, email, name, the <b>plain-text</b> API key                                                          |                                                                                                                                                       |
| <b>groups + groupmembers</b> | competition groups + who's in them                                                                             | <code>groupmembers</code> is a proper join table with a "one row per (group, user)" rule                                                              |
| <b>goals</b>                 | title, target hours, tech stack, deadline, status, <code>completedAt</code>                                    | a goal is marked complete from the Goals page (PATCH <code>/api/goals/:id/complete</code> , added 2026-09-10); before that nothing could complete one |
| <b>teams</b>                 | older idea; members stored inside the team record                                                              | backend exists but <b>the website never shows it</b> — dead code                                                                                      |
| <b>notifications</b>         | goal deadline reminders                                                                                        | created by the cron job                                                                                                                               |

### The `activities` record

Key fields: `userId` (stored as **text**, not a real reference — a quirk), `file/language/ project`, `duration` in **seconds** (added up as real measured seconds — the totals don't change from bucketing), line counts, `timestamp` (= the 10-minute window start), `files[]`, `flushCount`, and four sub-objects (`terminalAnalytics`, `editorAnalytics`, `focusAnalytics`, `gitAnalytics`) that now only store the numbers that actually happened (a dead `date` field was removed).

### Why MongoDB (a good answer)

- Activity records are **written once, never edited** (each 10-minute bucket is just added to with `$inc`) — good fit for a document store.
- The shape **grew over time** — I added three tracker sub-objects with **no migration**, because Mongoose just treats missing fields as zero.
- The main read is "give me one user's records for the last week, then add them up" — no joins needed on the hot path.
- Free managed hosting (Atlas).

### When SQL (Postgres) would be better

- The **relational bits** — group membership, goal ownership — where foreign keys and "delete cascades" would remove defensive code.
- The **leaderboard** — that's a `GROUP BY user` which SQL does with an index, instead of loading the whole table into Node.
- **Transactions** for multi-step actions.

## Indexes (updated 2026-09-08)

{userId, timestamp} (the one the read queries actually need — added), {userId, project}, {userId, language}, a **unique** {userId, project, language, bucketStart} (makes the 10-minute merge race-safe), and a **400-day auto-delete** rule on createdAt. The old {userId, date} index was dropped along with the date field. **Still missing:** anything that would help the global leaderboard — but that's a full table scan by design; the fix is a running-totals table, not an index.

---

## 7. The backend and API

### How one request flows

**Extension sending data:** POST /api/extension/track → check the API key → basic checks (file, language, duration present) → clean the numbers → work out which 10-minute window this belongs to → **add the numbers into that window's record** (findOneAndUpdate with \$inc, create it if missing) → return 201. (Set ACTIVITY\_BUCKET\_MS=0 and it goes back to one new row per flush.)

**Website reading data:** GET /api/analytics/:userId → check the JWT cookie → **check "is this your own data?"** (403 if not) → Activity.find(last 7 days) → add it up in JavaScript → return JSON → Chart.js draws it.

### The main endpoints

| What                  | Endpoint                                         | Login                                | Notes                                                                                       |
|-----------------------|--------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------|
| Send activity         | POST<br>/api/extension/track                     | API key                              | adds into the 10-minute bucket (one record per window, not per call)                        |
| Verify key            | GET<br>/api/extension/verify                     | API key                              | used during setup                                                                           |
| Daily / weekly charts | GET<br>/api/analytics/:userId<br>/weekly/:userId | JWT + ownership check                | adds up in JavaScript                                                                       |
| Insights              | GET /api/metrics                                 | JWT                                  | <b>no :userId in the URL</b> — uses your session only, so it can't leak someone else's data |
| Leaderboard           | GET /api/leaderboard                             | JWT                                  | reads the whole activities table every time                                                 |
| Groups                | /api/groups/*                                    | JWT (+ membership check for details) |                                                                                             |
| Goals                 | /api/goals/*                                     | JWT (+ owner check)                  |                                                                                             |
| Notifications         | /api/notifications/*                             | JWT                                  | correctly scoped to you                                                                     |
| Google login          | /auth/google,<br>/auth/google/callback           | —                                    | sets the JWT cookie                                                                         |

## Weak spots in the API

- Every route does its own `try/catch` and returns `error.message` to the client — that used to **leak internal details** — ☒ fixed 2026-09-09 with one central `(err, req, res, next)` handler (correlation id, generic body).
  - Rate limiting on `/auth` (50/15min) and `/api/extension` (120/min) since 2026-09-09, extended 2026-09-12 to group `/join` (10 per IP per 15 min) and `/api/analytics` (120/min).
  - Security headers via `helmet()` (added 2026-09-09).
  - Input validation since 2026-09-09: `duration` must be 1–3600 s, strings are length-capped, and `timestamp` must be within the last 24 h. The client still chooses the timestamp, and nothing caps how much time lands in one 10-minute window (H-21).
  - No pagination on the leaderboard.
- 

## 8. The frontend (React)

- **React 19 + Vite + TypeScript + Tailwind.** `react-router-dom` for pages.
- `App.tsx` checks `GET /api/user/profile` on load. If that works, show the app. If not, show the login page.
- **State:** just `useState` in each page, plus passing the `user` object down as a prop. The only React Context is the **theme** (28 colour themes, saved in `localStorage`).
- **No data-fetching library.** `@tanstack/react-query` is installed but **never used**, so every page reload re-fetches everything. **BETTER:** actually use React Query.
- **Charts:** `Chart.js` (Line, Pie, Bar).

### Things to know (and own honestly)

- `npm run build` is **green** as of 2026-09-09 — the TypeScript check had 29 old errors (mostly unused imports). `vite build` alone works. It's a cleanup job, not a design flaw.
  - The dashboard's **"Repeated Failures" panel** now shows the real `repeatedFailedCommands` (fixed 2026-09-09; it used to render fake hard-coded data).
  - The half-built **Goals to-do list was removed** (2026-09-09); goals themselves are saved.
  - The **Teams page isn't linked** anywhere.
- 

## 9. Insights — "is this machine learning?"

**No.** Say it plainly. It's plain statistics, calculated fresh every time you open the page.

### The metrics (simple definitions)

| Metric          | What it means                                         | How it's worked out                                          |
|-----------------|-------------------------------------------------------|--------------------------------------------------------------|
| Deep work ratio | how much of your real work time was in long stretches | minutes in blocks $\geq 25$ min $\div$ minutes in all blocks |
| Flow blocks     | the shape of your sessions                            | median, longest, count and total of your work stretches      |

| Metric                                               | What it means                                         | How it's worked out                                                                                         |
|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Volume stability</b> ( <i>was "consistency"</i> ) | how even your daily amount is, on days you code       | $1 - (\text{median absolute deviation} \div \text{median})$ of daily minutes — one huge day can't wreck it  |
| <b>Coding cadence</b>                                | how many days you code at all                         | active days $\div$ days in the window                                                                       |
| <b>Quality streak</b>                                | days in a row with real focus                         | consecutive days with at least one 25-minute block                                                          |
| <b>True peak window</b>                              | your <i>most productive</i> 2 hours, not your busiest | the 2-hour window with the most "surviving minutes" (minutes $\times$ (1 – churn)), seen on at least 3 days |
| <b>Estimation accuracy</b>                           | do you underestimate goals?                           | median of (hours actually spent $\div$ hours estimated) over completed goals                                |

**Count them as 11.** The seven above, plus four simple ratios: churn (lines rewritten  $\div$  lines typed), comprehension load (reading time  $\div$  reading + writing time), context switches per hour and interruptions per hour. Use "11" everywhere, resume included.

## Key points for the interview

- **Runs on every page load.** Only your 90-day baseline is cached (refreshed at most once a day).
- **Every number has a confidence level.** With too little data it shows "—" instead of a fake "0%" — three days of data never looks like six months.
- **"What stands out" is a rules engine, not AI.** 13 plain threshold rules read the numbers and say what's worth noticing (e.g. "you're rewriting more than usual"), each showing the numbers it fired on. A rule is skipped when its data is too thin.
- **Why statistics instead of ML?** No labelled outcomes to learn from, not enough users, and every number needs to be explainable. Statistics work from day one.
- **The AI part is designed but not built.** There's a plan for an LLM that would *describe* these numbers in sentences — with a strict rule that it can't invent any number — but no code for it yet.

## If asked "how would you add real ML?"

There's a written plan (`docs/ML_INTEGRATION_PLAN.md`), designed but **not built**:

- **What:** k-means clustering to learn session types instead of hand-written rules.
  - **Which numbers:** 8 that describe a session's *style* — how much code you rewrote, reading vs typing, build and command failure rates, deep-focus share, typing speed, terminal use, and file jumping. Session **length and commit count are left out** on purpose: they describe *size*, so the model would just sort long sessions from short ones.
  - **How:** train on a laptop in Python, save only the group centres, and label live sessions in JavaScript by nearest centre.
  - **How you know it works:** there's no answer key, so use the silhouette score and check how often it agrees with the existing rules.
  - **Why it isn't built:** only 4 usable sessions exist. Say that plainly — it's a better answer than a model nobody can inspect.
-

## 10. The social features — groups (the core) and the global leaderboard

**Groups are why the project exists** — friendly competition in a friend group. You make a group for a contest week or daily practice, everyone joins, and because everyone's editor is tracked automatically the group page ranks who actually did the work.

### The global leaderboard — how ranking works (REAL)

Rank = **total coding hours, all time, highest first**. Ties keep their existing order.

#### How it's calculated

Every time someone opens the page:

1. Load **every user**.
2. Run an aggregation over the **entire activities collection** (or an optional `?days=` window) — sum hours, lines and **real git commits** per user.
3. Merge and sort **in Node**.
4. Give each user scores (speed, quality, etc.) **relative to the current top user**, clamped to 0–5 — so everyone's score shifts when the leader codes more.

#### The problems

- **It reads the whole activity table on every request.** No time window by default, no cache, no pagination. This is the **first thing that breaks** as data grows.
- ~~It returns every user's email~~ — fixed 2026-09-12; it returns the display name only.
- *Fixed 2026-09-10:* "Commits" used to be the upload count, not real git commits; scores could go negative; and a `Math.max(...)` over every user would crash past ~100k users.

#### The fix (BETTER) — know this well

1. Keep a small **UserStats table**: one row per user with running totals, updated with `$inc` whenever activity comes in.
2. The leaderboard becomes `UserStats.find().sort().limit(50)` — reads 50 rows, not millions.
3. For "what's my rank?" instantly, use a **Redis sorted set**: `ZADD` on update, `ZREVRANGE` for a page, `ZREVRANK` for your position — all  $O(\log n)$ .
4. Add time-windowed tables for "this week" boards. Paginate. Cache the top 50.

#### Groups

- **Create**: name, description, public/private. Private groups get a **hashed** (scrypt) password. The creator becomes the first member.
- **Join**: public = one click; private = enter the password (old plain-text ones still work and get upgraded to a hash on the next correct login).
- **Membership** is a `groupmembers` join table with a "one row per (group, user)" unique rule — a solid race protection.
- **View details**: members-only (403 otherwise), then a **per-member leaderboard**: hours, lines, commits, failed commands and failed builds with failure rates (same slow full-scan pattern as the global one).
- **Leave**: the last member leaving deletes the group.

**The original idea — now built (2026-09-10):** the group table shows **how many errors each person hit** — failed commands and failed builds, with failure rates. The extension had always collected it; it was a read-path change (extend the same `$group` with `$sum` of the failure counters). A rate shows "—" when someone never ran a build, so "never failed" and "never ran" don't look the same. **Still open:** scoping the board to a contest week (`?from=&to= + a $match on timestamp`).

**Other group weak spots:** no admin/owner powers (`createdBy` stored but unused — no kick or rename); a double-join returns **409** (fixed 2026-09-09; the handler detects `11000`); "discover" lists private groups too (password-gated, not hidden — a deliberate choice; its search is escaped since 2026-09-10 — it was a ReDoS risk). Join is rate-limited to 10 tries per IP per 15 minutes (2026-09-12), but per IP, so a campus on one Wi-Fi shares it (H-20). Group pages still return emails (M-28, M-29).

## 11. Login and security

### The website login flow

1. Click login → go to `/auth/google` → Google → back to `/auth/google/callback`.
2. The server finds or creates your user, then signs a **JWT** with your ID, name, email. Valid for **1 day**.
3. The JWT goes into an **httpOnly cookie** (JavaScript can't read it).
4. Every protected route reads the cookie, verifies the JWT, loads the user.
5. Logout clears the cookie.

### Login weak spots

- The signing secret is required at boot — the app refuses to start without `JWT_SECRET` (fixed 2026-09-09; was an unsafe fallback).
- **No refresh token** — after 1 day you're just logged out.
- **No way to revoke** a token early (no server-side session list).
- The cookie is `sameSite: 'none'` in production with no CSRF token — low impact here because most actions are POSTs with a JSON body, but worth mentioning.

### Security — already fixed (on the current branch)

- Analytics endpoints now check **"is this your data?"** (was: anyone with your ID could read your history).
- Old **unauthenticated** write/read endpoints **deleted**.
- Group passwords are now **hashed** (were plain text).
- The `AUTH_BYPASS` dev flag is **refused in production**.
- Goal and team endpoints now check ownership/membership.
- There's a test that **statically scans the routes** and fails if any protected route loses its auth middleware.

### Security — still open (be honest about these)

| Issue                                                                                                                                                                             | Why it matters                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| API key is plain text, never expires, no limits                                                                                                                                   | leak = someone fakes your activity                                                            |
| Rate limiting only on <code>/auth</code> + <code>/api/extension</code> <br>extended 2026-09-12 | group <code>/join</code> now 10 tries per IP / 15 min;<br><code>/api/analytics</code> 120/min |

| Issue                                                                                                  | Why it matters                                                             |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| No security headers <input checked="" type="checkbox"/> helmet() added 2026-09-09                      | HSTS, nosniff, no X-Powered-By                                             |
| No input validation on activity <input checked="" type="checkbox"/> 2026-09-09                         | duration capped at 3600, length/timestamp bounds; still no idempotency key |
| Errors return error.message <input checked="" type="checkbox"/> fixed 2026-09-09                       | central handler, generic body + id                                         |
| Leaderboard shows every email <input checked="" type="checkbox"/> fixed 2026-09-12                     | name only now                                                              |
| Group pages show members' and creators' emails (M-28, M-29)                                            | privacy; same class as the leaderboard leak                                |
| No duplicate/replay protection                                                                         | resend a request → counted again                                           |
| Group search puts user text straight into a regex <input checked="" type="checkbox"/> fixed 2026-09-10 | input is escaped and length-capped                                         |

"Can users cheat the leaderboard?" — yes, easily. curl the track endpoint with duration: 3600 in a loop using a valid key; one call = one fake hour. Fix: validate the duration (1–3600s), rate-limit per key, require a unique ID per send, and only trust hours that also have editor/focus/git activity.

## 12. Scaling up — 10k, 100k, 1M users

| Users     | What happens                                                                                                                                                                           | What to do                                                                                                                                                                                                                                             |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10,000    | Mostly fine. Writes are now ~10× lower (10-minute buckets), and the timestamp index + 400-day auto-delete are in. The leaderboard still gets slow once activities is millions of rows. | Repoint the leaderboard / all-time reads at the dailysummaries rollup that already exists; put a 30-day window + limit on the leaderboard query.                                                                                                       |
| 100,000   | Leaderboard is unusable without a running-totals table. No caching means the same maths runs over and over.                                                                            | Build the UserStats rollup (\$inc on write). Move the JS-side analytics into MongoDB \$group.                                                                                                                                                          |
| 1,000,000 | One server + one database is the wrong shape.                                                                                                                                          | Put a queue in front of activity ingest → workers write to a sharded activities (split by userId) plus rollups. Redis for the leaderboard and the insights cache. Serve charts from precomputed rollups. Archive old raw records. Add real monitoring. |

**The one-line answer:** "Yesterday's change — merging flushes into 10-minute buckets on write — cut the write and row rate about 10× with no change to the numbers. The next bottleneck is still the leaderboard, because it reads the whole activity table per request; the fix is a per-user running-totals table updated on write, turning a full-table scan into a 50-row read, and I already built the daily-summary collection it would sit on."

## 13. Testing and deployment

### Testing — the honest picture

**What exists:** small test scripts using Node's built-in `assert` (no framework).

- **Backend:** 18 files, 296 assertions (re-run 2026-10-03) — the streak logic, the number-cleaning functions, the insight formulas, the rules engine, the leaderboard scoring, the ownership helper, the password hashing, the route scanner that enforces auth, and (2026-09-08) the bucketing maths, the schema/index shape, the ingest wiring, and the daily-rollup builder.
- **Extension:** 3 files, 52 assertions — the trackers' behaviour, an "activation" test that loads the real built bundle and checks every command is registered and no network call happens without a key, and a flush-safety test for the carry-forward of unsent data.

**What's missing:**

- **No integration tests** — nothing starts the server and hits a real database.
- **No frontend tests at all.**
- Nothing has ever run against a real database — the queries are only reasoned about.

**BETTER:** add `supertest` + an in-memory MongoDB for API tests; React Testing Library for the frontend; one end-to-end test with Playwright.

### Deployment

- **Website:** Vercel. Build is `tsc -b && vite build` (✅ green as of 2026-09-09).
- **Backend:** Render (a normal always-on Node process). There's also a Vercel serverless config in the repo.
- **Database:** MongoDB Atlas.
- **Extension:** packaged with `vsce`, published to the VS Code Marketplace.
- GitHub Actions CI on push (2026-09-09): backend + extension tests, frontend build. Still no CD, no Dockerfile.

**One gotcha:** the deadline-reminder cron job uses `node-cron`. That works on Render (always on) and — since 2026-09-09 — is serverless-safe: `initScheduler()`/`app.listen()` are behind a `require.main` guard, and an external trigger hits `POST /api/internal/run-*`. Before the fix the hourly job never fires. **BETTER:** use Vercel Cron or an external scheduler hitting a protected endpoint.

---

## 14. Quick Q&A (memorise these)

**Why MongoDB?** Activity records are write-once and the shape changed over time — I added three sub-objects with no migration. Main read is "one user's week, then sum". No hot-path joins.

**Why not Postgres?** Honestly, the relational parts (groups, goals) and the leaderboard would be better in SQL. The activity stream suits documents. If I redid it with the social features front of mind, I might pick Postgres.

**Why a monolith?** One developer, one database, shared data model. I'd split out the activity-ingest endpoint first when volume demands it.

**Is the API key authentication?** Yes — it's a bearer credential, not an identifier. Plain text, no expiry, no scope.

**What if someone steals a key?** They can fake your activity and mess up the leaderboard. They can't read your dashboard — that needs the login cookie.

**How would you fix the key?** Hash it at rest with an ID prefix, allow multiple per user, rotation with a grace period — or move to short-lived tokens / OAuth device flow.

**Why not OAuth for the extension?** A device flow is the right answer; the key was faster to ship. It's a legitimate MVP trade-off.

**Why JWT and not sessions?** Stateless — no session store to run, survives restarts. The cost is no easy revocation and no refresh token.

**How does data get from the extension to the dashboard?** Extension event → tracker counters → ~2-min flush → check API key → `$inc` the numbers into the right 10-minute record → (later) website asks → check JWT + ownership → read records → add up → JSON → chart.

**Where's the aggregation done?** Mostly in JavaScript after `Activity.find()`. Tech debt — it should be MongoDB `$group`. Streak, insights, the group leaderboard and the daily rollup already use `$group`.

**What's the bottleneck?** Still the leaderboard — it reads the whole activity table on every request. (Yesterday's 10-minute bucketing cut the row count ~10×, buying time, but the complexity class is unchanged.)

**How do you scale the leaderboard to 10M users?** A per-user `UserStats` rollup updated on write, then a Redis sorted set for  $O(\log n)$  rank lookups.

**Is Insights machine learning?** No — plain statistics (averages, medians, a spread score), computed fresh each load. An LLM description layer is designed but not built.

**Why statistics not ML?** No labelled data, not enough users, and every number must be explainable.

**Can users cheat the leaderboard?** Easily — send fake hours with a valid key. There's no rate limit or validation. Fix: validate duration, rate-limit, require a unique send ID, cross-check with editor activity.

**What happens if the same activity is sent twice?** It `$incs` the same 10-minute record again — so it double-counts *within* that bucket, but doesn't add a phantom row. Still not fully idempotent; proper fix is a client-generated ID per send + a unique index.

**Two requests at the same time — any problem?** Ingest is now an atomic `findOneAndUpdate` with `$inc` — two flushes for the same window both apply cleanly, and the unique bucket index handles the create race with a retry. The weak spots elsewhere are group-join (unique rule catches it but returns a 500) and adding a team member (a read-modify-write — should be `$addToSet`).

**Why is `userId` a string on activities but an ID everywhere else?** Historical. It forces type-conversion tricks in the leaderboard and blocks joins. Migrating it is on the list.

**What breaks first at scale?** The leaderboard (whole-table scan per request), then the serverless cron. The per-user analytics scans are now indexed and bounded by the bucketing.

**Your frontend build — walk me through the cleanup.** It had 29 old TypeScript errors (mostly unused imports, plus 5 implicit-any props in one decorative component). Fixed 2026-09-09; `npm run build` is green

and CI keeps it that way. It was a cleanup, not a design issue.

**How do you know the database queries are correct if nothing ran against a real DB?** I don't fully — the pure functions are unit-tested (and the streak test runs the *shipped* code, not a copy), but there's no integration test. That's the biggest gap and it's on the plan.

**What was the hardest part?** Making the *time* number honest — idle time, and lines that net to zero on a refactor. I built an idle/pause state machine and switched to gross edit counts plus a churn metric.

**What would you improve first?** Hash the keys; add the error/build-success columns to the group leaderboard (the original motive — read-path change only); repoint the all-time reads at the daily-summary rollup that already exists; fix the TypeScript errors; add integration tests.

**You changed the write model recently — what and why?** It was one row per flush (~1 per minute of coding). I changed ingest to `$inc` into a 10-minute (`user`, `project`, `language`) record. `$inc` adds the real measured seconds, so every total is identical — only the time-of-day detail is now 10-minute-grained, which is the finest any chart shows. Row count dropped ~10×. There's a rollback env switch (`ACTIVITY_BUCKET_MS=0`).

---

## 15. "Defend your project" — the tough follow-ups

**"Why MongoDB?" → "But you have a join table and embedded arrays — isn't that SQL fighting a document store?"**

Partly true. The join table is me reaching for a relational pattern because groups need it, and it works — the unique index gives the guarantee. It's a sign the *social* half would be happier in Postgres. The *activity* half, which is the bulk and the hot path, genuinely suits documents.

**"Your leaderboard is O(n) per request." → "A cache fixes that."**

A cache hides the cost behind a timer — the full scan still happens. The real fix is the rollup table, which changes the complexity. A cache on top of that is a nice extra.

**"Why not Redis / a queue / WebSockets?"**

Nothing needs them yet. Leaderboard and insights finish in under a second at demo scale. Redis is the obvious next dependency (leaderboard + cache + rate limits). A queue earns its place past ~10k concurrent users. WebSockets I'd add for real-time group activity, not for a 30-second notification poll.

**"You could have shipped faster with Firebase."**

True for the CRUD and auth. But the analytics are custom aggregations over a high-volume collection, and Firebase's per-document-read pricing would hurt — the daily chart alone reads a week of records. And the extension needs a stable server API regardless.

## "How do you stop cheating if a determined user spreads fake data thinly?"

Cross-check signals. A real coding hour has editor edits and focus events. Weight the leaderboard on *corroborated* activity, and cap how much any single send can contribute.

## "Nothing is tested against a real database — how do you trust it?"

I trust the pure logic (unit-tested, and the streak test runs the real shipped code). I don't fully trust the queries end-to-end — that's stated in my notes. `supertest` plus an in-memory MongoDB closes it and it's the top testing item.

## "If this got 10,000 real users tomorrow, what's your first move?"

Add the `timestamp` index and put a 30-day window on the leaderboard query as a stopgap, then build the `UserStats` rollup. In parallel: turn on `helmet` and rate limiting so ingest can't be flooded, and add validation so `duration` can't be faked. Then a health check and real logging.

## 16. Final reminders

### The weaknesses to own before they're pointed out (updated 2026-10-03)

#### Still open — say these yourself:

1. **Cross-site login cookie (H-19)** — the site and the API are on different domains, so Safari, Firefox and incognito probably block the login cookie. Fix: serve the API through the frontend's domain (Vercel rewrites) or a custom domain.
2. **Rate limits are per IP (H-20)** on sign-in, analytics and group join — a campus on one Wi-Fi shares them. Fix: limit per user after authenticating.
3. **Client-chosen timestamps** — bounded to the last 24 h, and credited time is capped, but a script that fakes focus and paces itself still earns an hour per real hour.
4. **No refresh token** for the web login (1-day JWT, no server-side revoke).
5. **Windowed boards** (`?days=`, contest weeks) still sum raw activity in the window; all-time boards use running totals.
6. **No frontend tests.**
7. **Not deployed yet** — the October work is built and tested; deployment and live migrations are pending (`docs/RELEASE.md`).

**Fixed in October 2026 (be ready to explain how):** plaintext API keys → hashed + per-device sign-in; full-scan leaderboard → running totals (6.7 s → 62 ms at 1M rows); JS analytics → `$facet` pipeline; no integration tests → 35; memory-only offline buffer → persisted outbox; replay double-counting → idempotency key; leaderboard inflation → credited time with a window cap; dead Teams code → deleted; all-time-only group boards → contest windows; browsers blocking PATCH (H-23) → fixed.

For each one: know *why it matters*, *what it would take to fix*, and *why it's acceptable for a student project right now*.

## Don't trip on these

- Don't call Insights "AI" or "ML". It's statistics plus rules. Say so first.
- Don't say anything from October 2026 is "in production" until it's deployed.
- Don't say the leaderboard can't be gamed — it's bounded, not impossible.
- Don't quote "~10× fewer writes" — measured is ~5× fewer documents (13× less data) at the 2-minute cadence; always add "local benchmark".
- Don't claim frontend tests. Integration tests exist now (35, in-memory MongoDB).
- The ML work-type classifier is designed, not built.
- There are 3 contributors — you owned the extension and the backend.

## Five sentences that make you sound senior

1. "The API key is basically a plain-text password with no expiry — a fine MVP, but I'd hash it at rest with an ID prefix and add rotation with a grace window."
2. "The leaderboard reads the whole activity table per request; the fix is a per-user stats rollup updated on write, turning a full scan into a 50-row read, with a Redis sorted set for rank lookups."
3. "I recently moved ingest from one row per flush to an atomic `$inc` into a 10-minute `(user, project, language)` bucket — real seconds are summed either way, so every total is identical, but the write and row rate dropped about 10×."
4. "The project's really about friendly competition — the group leaderboard is the point, and the next feature is surfacing the per-person error and build-success comparison, which the extension already collects."
5. "I audited my own code into an improvement plan — about 30 findings by severity — and closed the serious security ones with a test that fails if a protected route loses its auth check."

## 17. Which college subjects this project actually uses

If someone asks "where did you use your core subjects?", answer with a file, not a course name.

| Subject  | Where                                             | Say this                                                                                                                                                                                           |
|----------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DBMS     | <code>models/Activity.js</code>                   | "Compound index in equality-then-range order; I checked with <code>.explain()</code> that it became an index scan. A partial unique index stops a double write."                                   |
| OS       | <code>app.js</code> , extension flush loop        | "Node is one event loop, so races happen at the database, not in memory — I fixed them with atomic <code>\$inc</code> . And a timer-based cron cannot work on a host that freezes the process."    |
| Networks | <code>app.js</code> , <code>routes/auth.js</code> | "CORS allowlist, <code>httpOnly</code> + <code>SameSite=None</code> cookies because the SPA is on another domain, per-IP rate limits, and <code>trust proxy</code> so the real client IP is used." |

| Subject               | Where                                  | Say this                                                                                                                                                            |
|-----------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theory of computation | <code>services/textQuery.js</code>     | "A regex is a backtracking automaton. User input went straight into one, so <code>(a+)+\$</code> could freeze the whole server. Escaping makes it a literal match." |
| Algorithms            | <code>routes/leaderboard.js</code>     | "The leaderboard is $O(\text{all activity} + \text{all users})$ per request. Also <code>Math.max(...array)</code> throws past ~100k arguments, so I fold instead."  |
| Statistics            | <code>services/metricsDerive.js</code> | "Median and MAD instead of mean and standard deviation, because one huge day shouldn't dominate. Every number carries a confidence level."                          |
| Security              | <code>services/passwordHash.js</code>  | "scrypt with a salt for group passwords, <code>timingSafeEqual</code> for the cron secret, and an honest description of what a stolen API key does."                |
| Software engineering  | <code>backend/tests/</code>            | "The maths is pure functions with no database, so 296 assertions run in a second. Some tests exist purely to encode a bug that already happened."                   |

**And be honest about the gaps:** no machine learning, no threads or locks, no consensus or sharding, no compiler work. Saying that unprompted lands better than padding the list.